DIGITAL PUBLIC INFRASTRUCTURE

Institutional Evolution and Technical Architecture of UIDAI

A Comprehensive Evaluative Framework of India's Digital Identity Ecosystem

Unique Identification
Authority of India

AADHAAR
मेरा आधार
मेरी पहचान

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E-Governance: Digital transformation of India's public sector has made e-governance primary connection between state and citizens.

Massive Scale: Scaling digital infrastructure, successfully meeting the demands of over 1.4 billion people.

Foundation of India Stack: Aadhaar (under MeitY) became the bedrock of the India Stack.

Unique Identification Authority of India

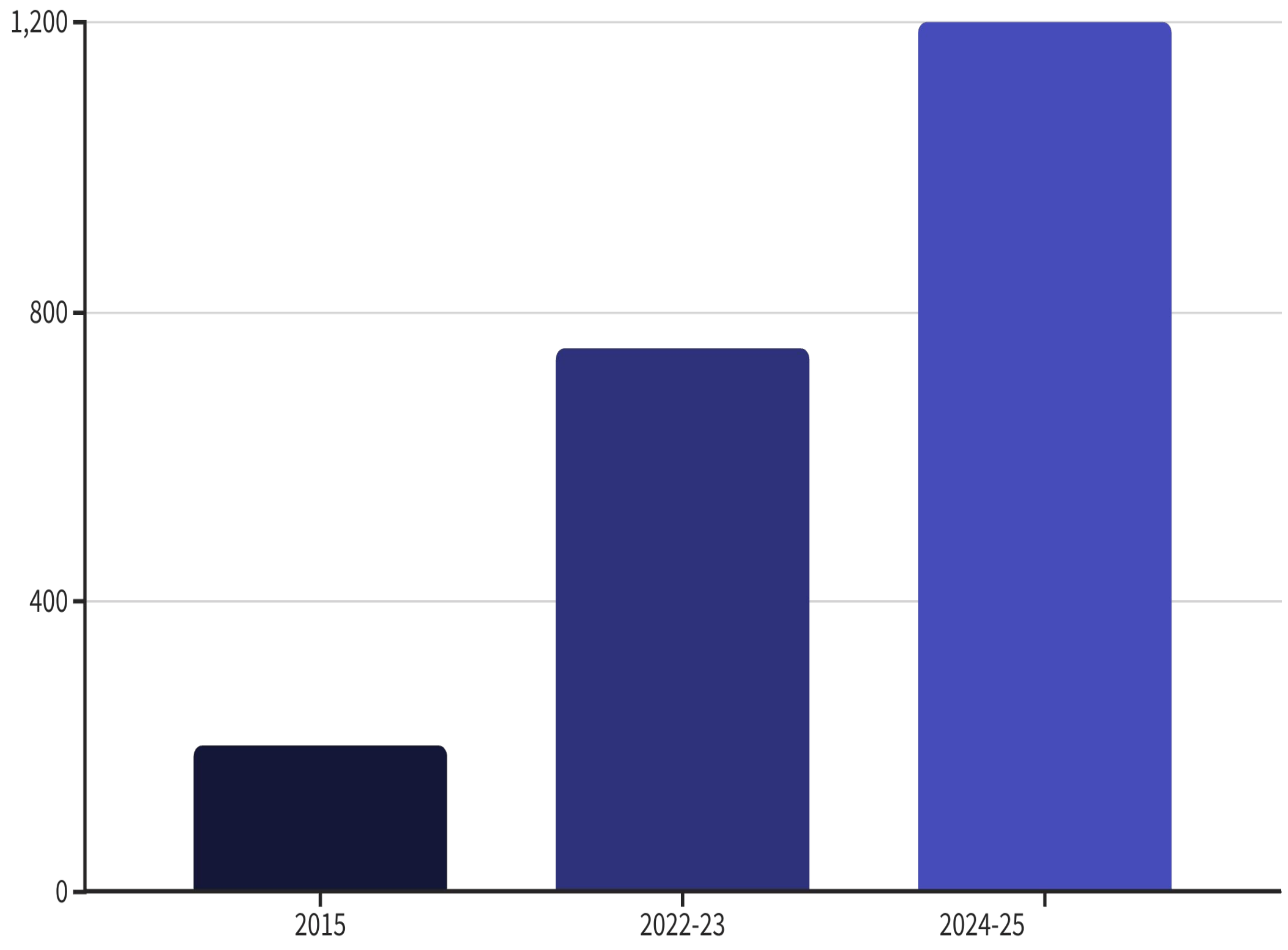
AADHAAR

मेरा आधार मेरी पहचान

📎 *Reference: [UIDAI Annual Report 2024-25]*

DIGITAL ECONOMY & DBT

Macroeconomic Context and Fiscal Governance



Key Fiscal Outcomes

GDP Contribution: India's digital economy contributed 11.74% to the national income in 2022–23 and reached to 13.42% in 2024–25.

Economic Expansion: Driven by e-governance, the digital economy grew massively from $200 billion in 2015 to an $1.2 trillion in mid-2025.

Government Savings: Aadhaar-linked Direct Benefit Transfers (DBT) successfully eliminated duplicate beneficiaries, resulting in savings of over ₹4.31 lakh crore between 2015 and 2025.

📎 *Reference: [PIB — Digital Economy Contribution to GDP 2024-25]*

INSTITUTIONAL ROBUSTNESS

From Project to Statutory Authority

1 **Planning Commission Era**

UIDAI began as an attached office of the Planning Commission, tasked with designing a national identity framework

2 **Aadhaar Act, 2016**

Parliamentary legislation transformed UIDAI into an independent statutory body under MeitY, granting strategic autonomy

3 **Scale Today**

The authority now supports over **142.76 crore** generated Aadhaar numbers, underpinning financial inclusion at national scale

📎 *Reference: [UIDAI — About the Authority]*

POPULATION-SCALE INFRASTRUCTURE

Technological Scaling and Authentication Capacity

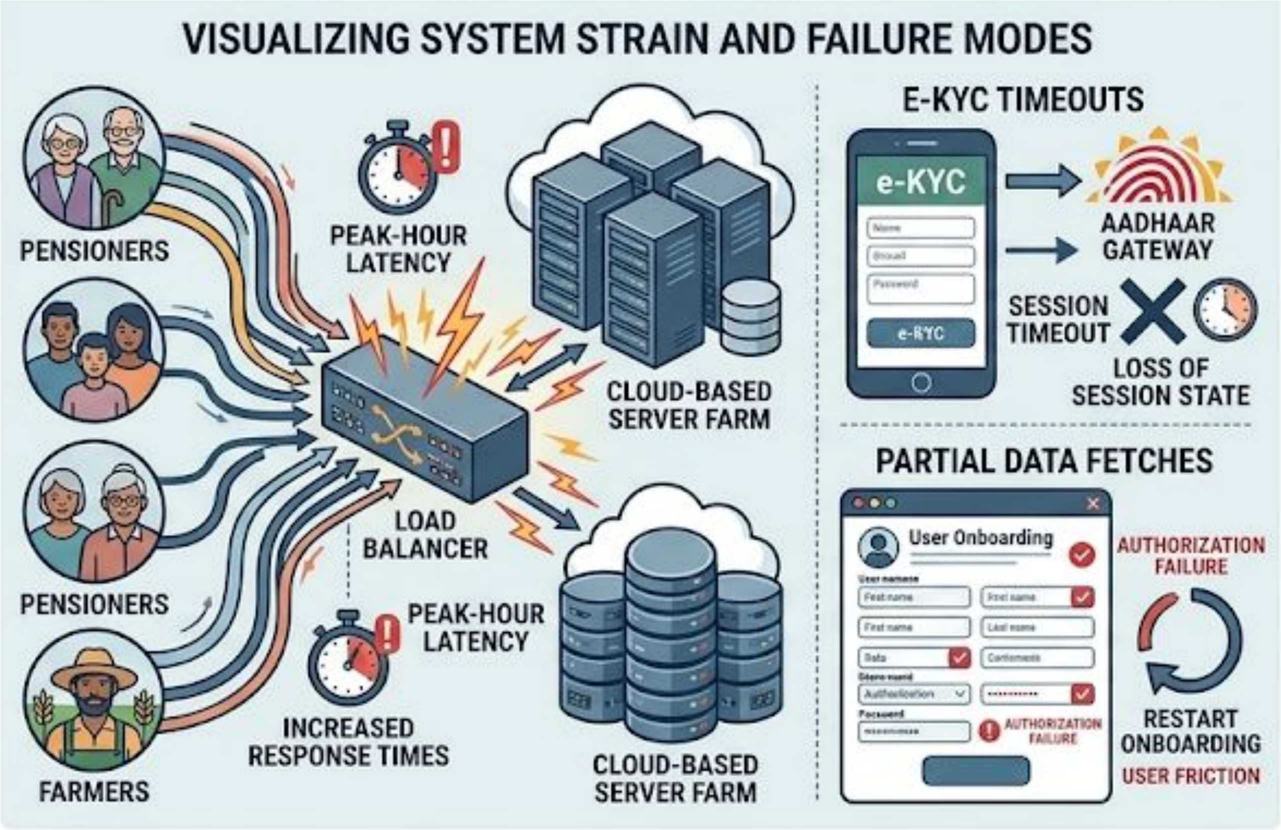
142.76Cr	231Cr	9.6Cr
Aadhaar Numbers Generated	**Monthly Transactions**	**Daily Average**
Total residents enrolled in the CIDR to date	Authentication requests processed in November 2025 alone	Authentications processed per day across all service channels

The Central Identities Data Repository (CIDR) transformed Aadhaar into a foundational identity platform for all residents. Infrastructure now features AI-driven face authentication capabilities alongside fingerprint and iris modalities.

📎 *Reference: PIB — 231 Crore Authentication Transactions, November 2025*

⚠ System Reliability

Technical and Session Management Vulnerabilities



Where the System Strains

Peak-hour latency: Simultaneous requests during pension renewals and crop subsidy disbursements trigger transaction failures across departments.

E-KYC timeouts: Users switching between service apps and the Aadhaar gateway frequently lose session state mid-flow.

Partial data fetches: Authorization failures force users to restart onboarding, compounding digital friction for low-literacy populations.

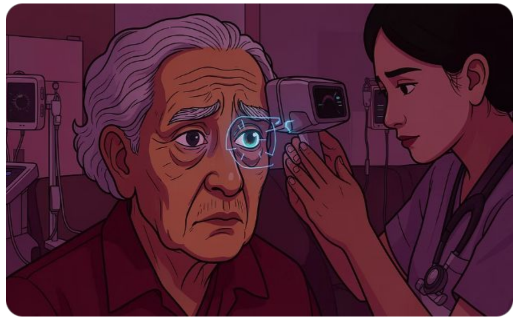
📎 *Reference: [AI CERTS — India's Biometric Governance Systems Surge]*

DIGITAL ACCESSIBILITY

Inclusivity Barriers for Diverse User Groups

Manual Laborers

Worn fingerprint ridges cause persistent "mismatch" errors, blocking access to wages and entitlements



Elderly Populations

Fading iris ridges and cataracts trigger scan failures, creating barriers to rations and pensions

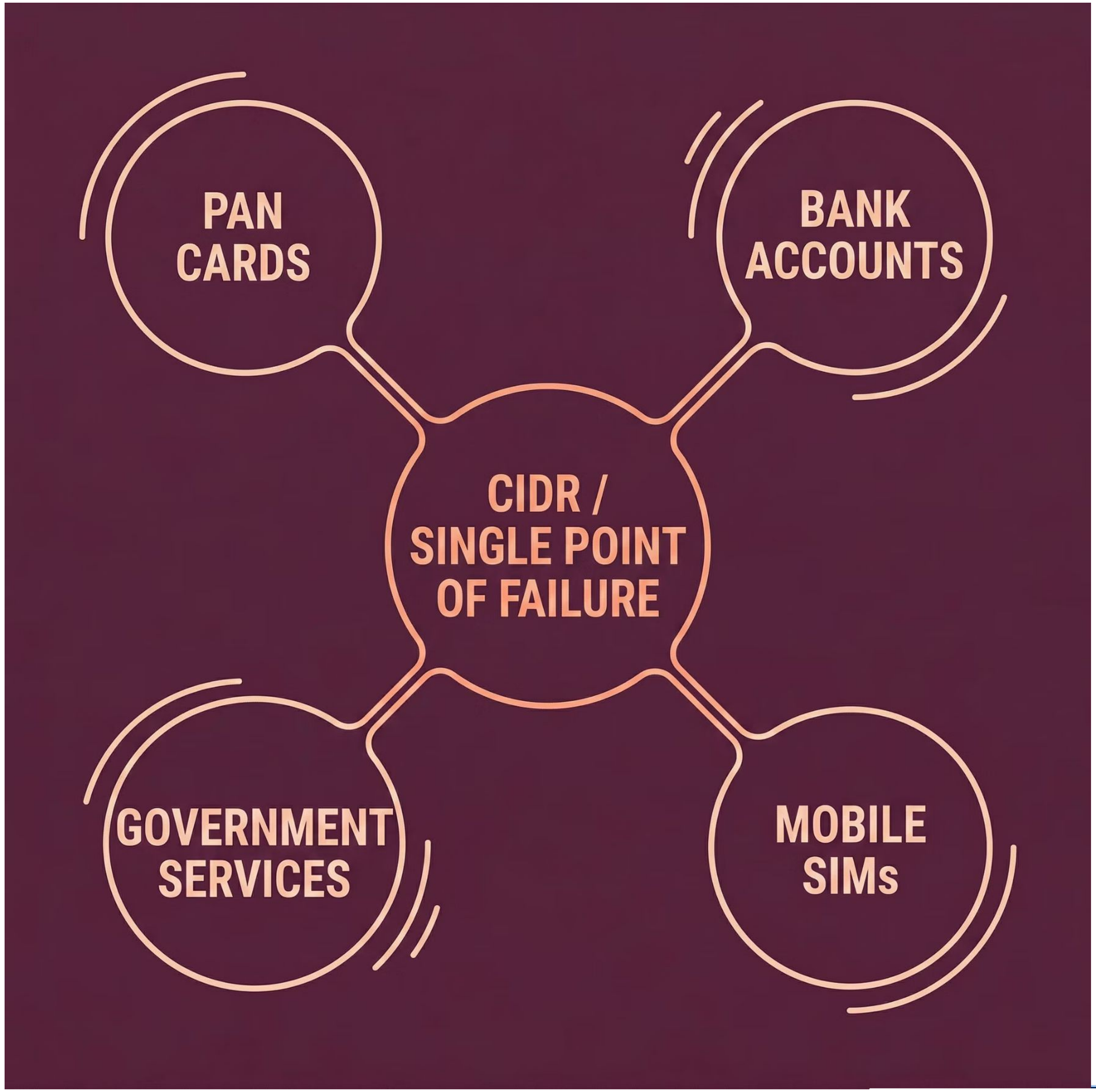
Persons with Disabilities

The Mandatory Biometric Update (MBU) process for Divyangjan remains physically convoluted and inaccessible

📎 *Reference: [WatchDoQ — Aadhaar Biometric Problems in 2025]*

🔒 Data Governance

Data Security and the Centralization Paradox



The Structural Risk

Honey-pot effect: The centralized CIDR is a high-value target — a single breach could compromise an individual's entire digital life
Compounded exposure: Linking Aadhaar with PAN, bank accounts, and mobile SIMs exponentially amplifies the impact of any vulnerability
Uneven compliance: "Aadhaar Data Vaults" were introduced, but implementation across thousands of AUAs remains inconsistent

📎 *Reference: JISA Softech — UIDAI 2025 Guidelines & Data Compliance*

E-SERVICE QUALITY & AUTOMATED SUPPORT

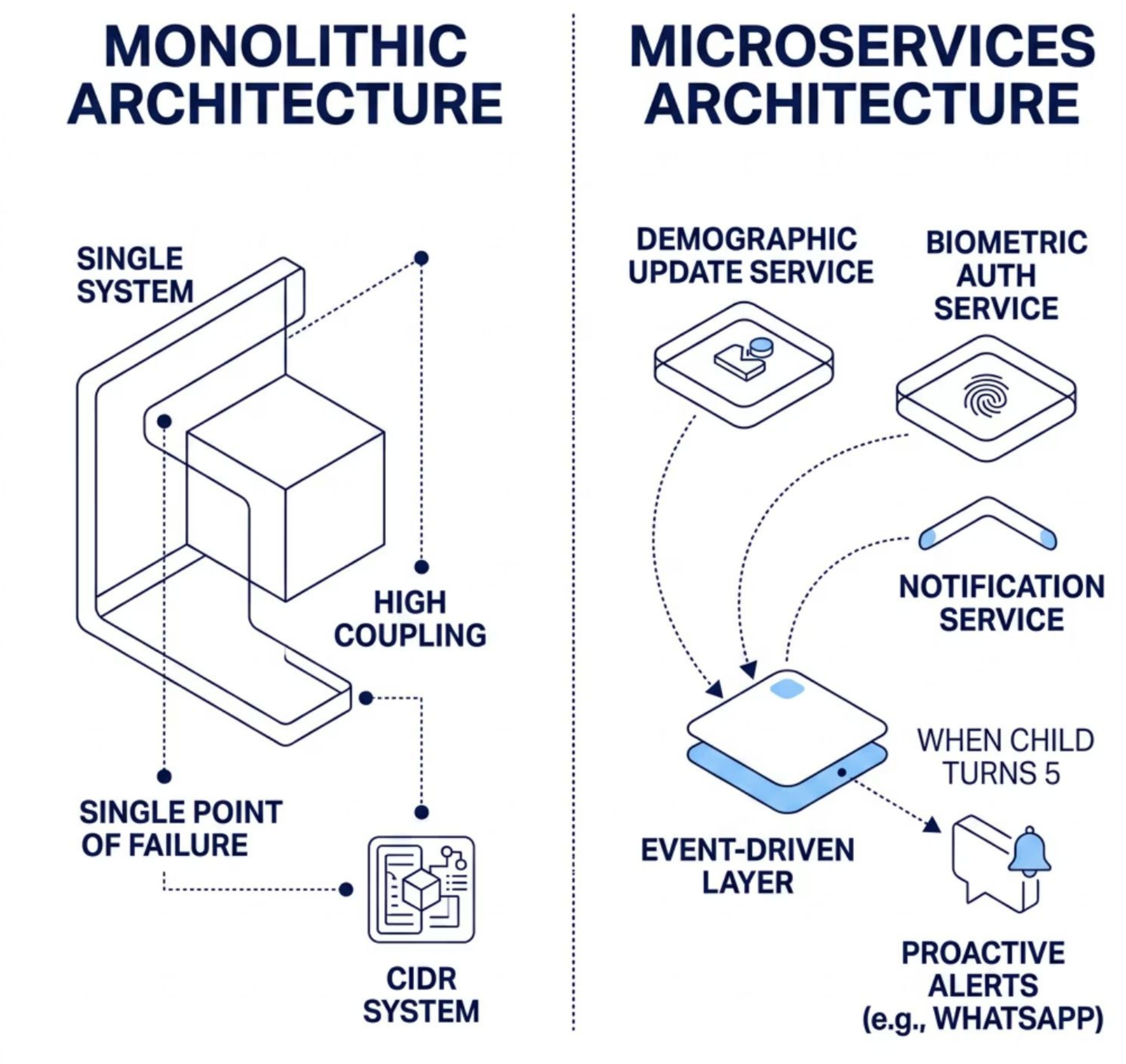
The Limits of AI in Grievance Redressal

UIDAI's **Aadhaar Mitra** chatbot resolves **~92% of standard complaints** within a week, offering 24/7 support. Yet its scope is narrow — it cannot initiate demographic updates or resolve biometric mismatches. Non-standard queries trap users in preset answer loops, ultimately redirecting them to physical enrollment centers.

Reference: [All you need to know about Aadhaar Mitra | The Economic Times]

AGILE ARCHITECTURE & FAULT ISOLATION

Transitioning to Microservices & Event-Driven Models



The Architectural Shift

Migrate from a **monolithic legacy framework** to a **Microservices Architecture**. Independent services — Demographic Update, Biometric Auth — prevent high load on one node from crashing the entire CIDR system.

Event-Driven Execution

Asynchronous communication enables **proactive governance** — automatically alerting parents via WhatsApp when a child turns 5 for mandatory biometric updates.

Reference: The 2025 Microservices Architecture Roadmap

CITIZEN-CENTRIC INTERFACE DESIGN

No-Document Updates and Automated Verification

Unified No-Document Flow

Replace manual uploads with **automated cross-checks** against linked databases (PAN, Passports) for single-tap demographic updates.

Multimodal Fallbacks

After **three fingerprint mismatches**, the system auto-defaults to Face Authentication or OTP — reducing citizen friction.

Session Persistence

Biometric-locked background sessions prevent timeouts during complex multi-step processes like banking onboarding.

Reference: *New Aadhaar Rules from November 2025*

OPEN NETWORKS & DIGITAL COMMERCE

Integrating with ONDC and the India Stack

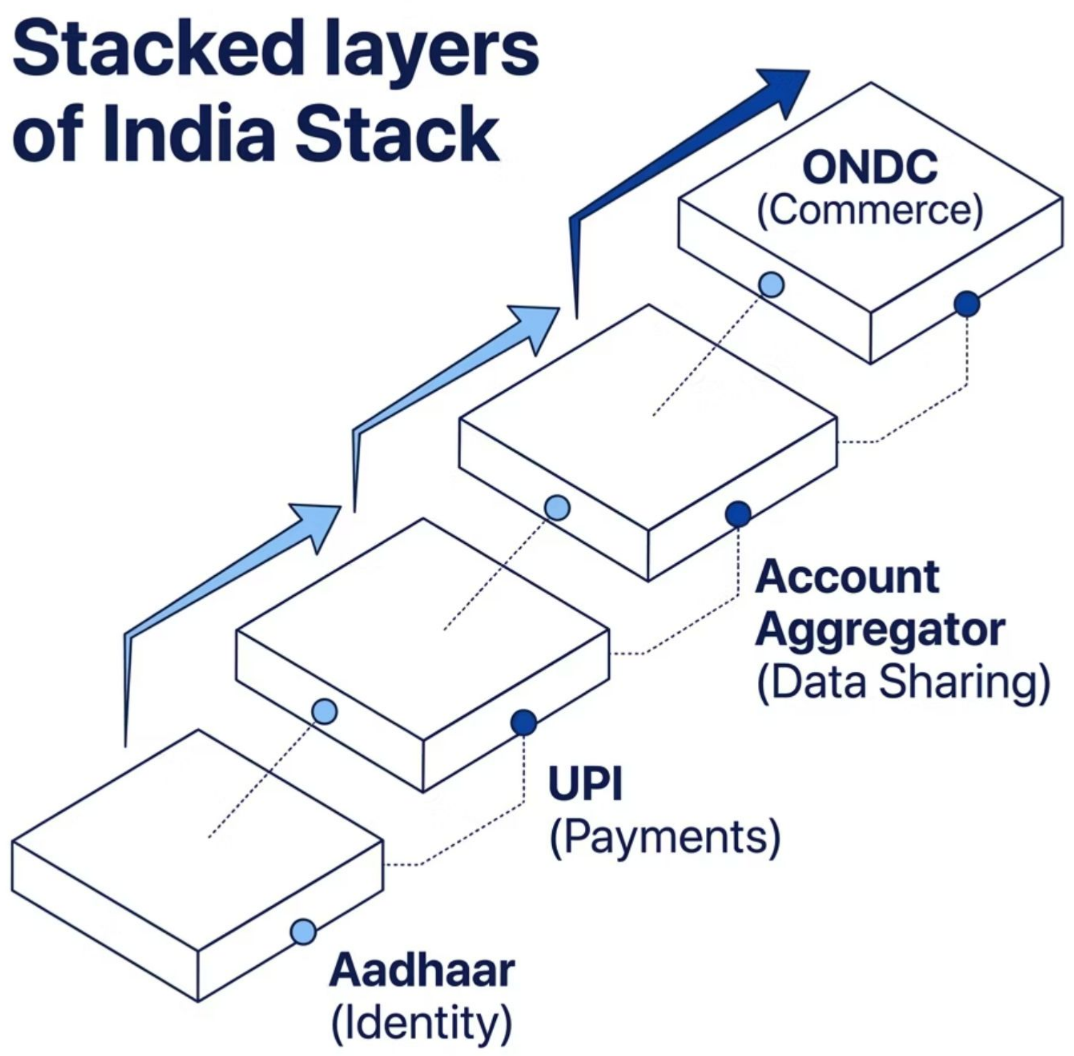
Commerce Layer Integration

Integrating Aadhaar with the **Open Network for Digital Commerce (ONDC)** slashes merchant verification costs from **₹20 to ₹0.15** — a 99% reduction enabling micro-merchants to participate in the formal economy.

Mobility & Transit

An integrated digital wallet could allow citizens to book a **single QR ticket** covering metro, bus, and long-distance trains — unifying identity with mobility.

Reference: India Stack — The Indian Approach and Experience



IS SUCCESS MODELING

Measuring E-Governance Information Systems Success

The **DeLone & McLean IS Success Model** evaluates Aadhaar across three critical dimensions of system effectiveness.

System Quality

High performance with **sub-second latency**, though flexibility remains limited for users in low-network or rural areas.

Information Quality

High biometric accuracy is a core strength. Upcoming document-free updates will significantly improve demographic data timeliness.

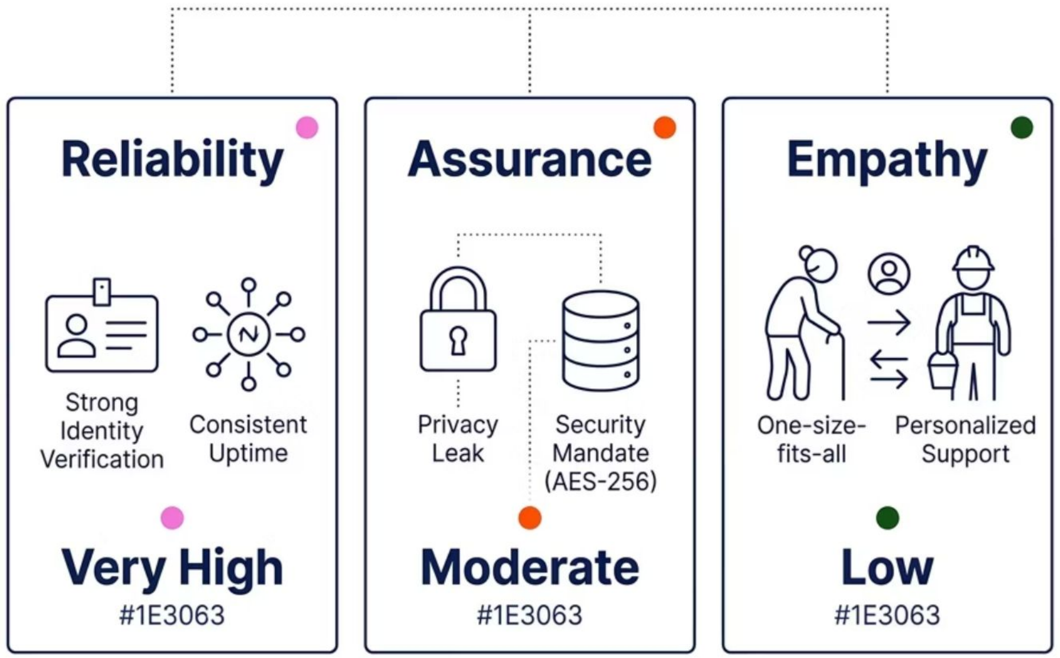
Net Benefits

Aadhaar acts as a **macroeconomic growth catalyst**, underpinning a **$1.2 trillion digital economy**.

Reference: Validation of the DeLone and McLean IS Success Model

DIMENSIONS OF SERVICE QUALITY

Assessing Citizen Satisfaction via SERVQUAL



Key Findings

- **Reliability:** Exceptionally strong in identity verification — Aadhaar's core strength.
- **Assurance:** Moderately impacted by privacy concerns; new **AES-256 encrypted vault** mandates are strengthening trust.
- **Empathy:** Currently the weakest dimension for elderly and laborers — requires a shift to **personalized, multi-modal support**.

Reference: [SERVQUAL — Measuring Consumer Perceptions of Service Quality]

FUTURE-PROOFING DIGITAL GOVERNANCE

Vision 2032 & Quantum Readiness

Immediate Reforms

Paperless updates and **free biometric windows for children** directly target the highest areas of user dissatisfaction identified through SERVQUAL.

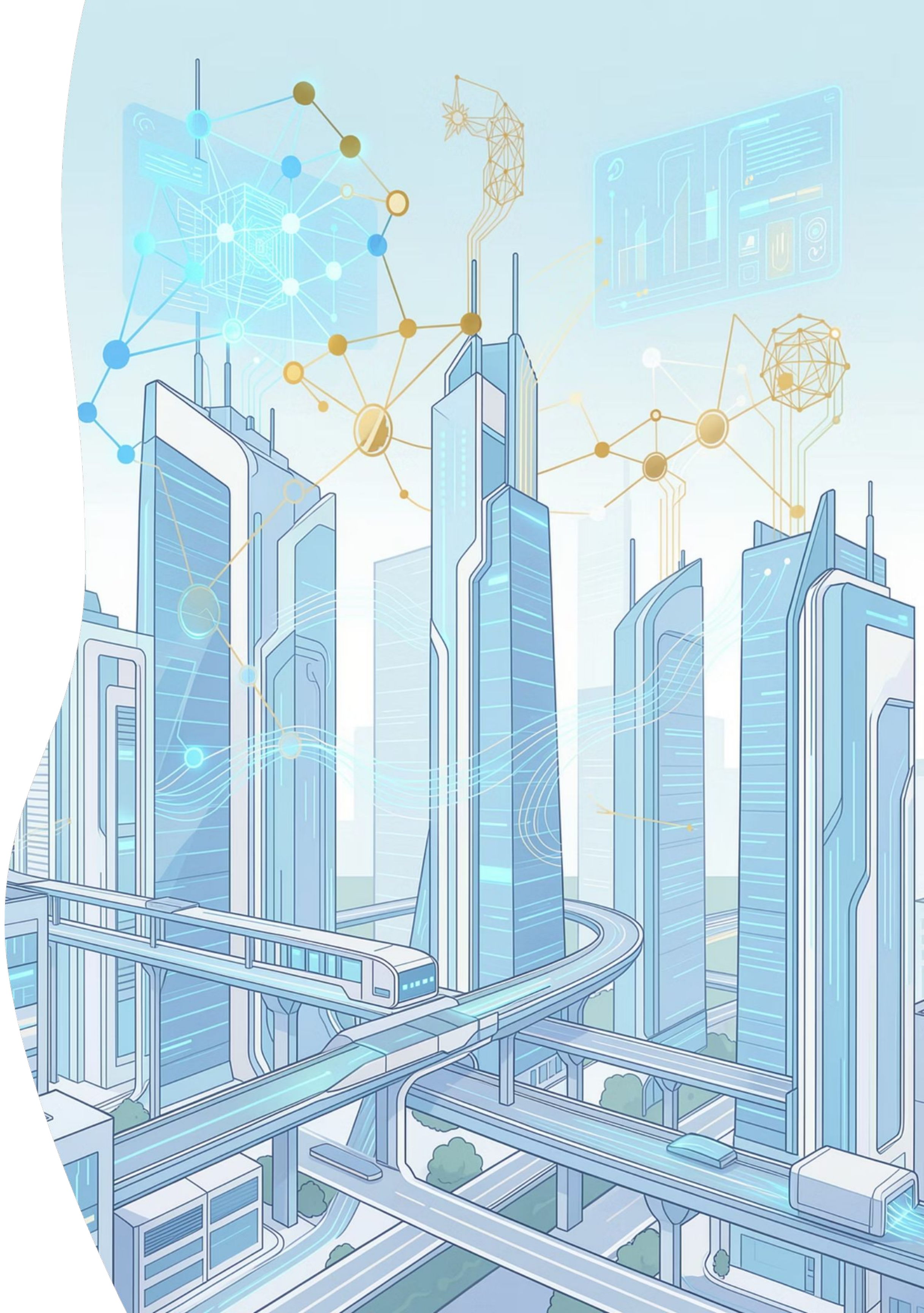
Policy Alignment

Aadhaar Vision 2032 will align with the DPDP Act and emerging global digital identity standards for cross-border interoperability.

Next-Gen Architecture

Future systems will leverage **blockchain** for decentralization and **quantum-resistant encryption** — ensuring a robust, people-centric digital identity.

Reference: [Aadhaar Vision 2032 Committee Launches]



Thank You

We are open to Questions ?